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Inventor(s)	Yu; Chen-Hua et al.

OPTICAL PACKAGE AND METHOD OF MANUFACTURE

Abstract

A method includes forming a laser diode structure including an active layer sandwiched between an n-type contact layer and a p-type contact layer; forming an n-type contact on the n-type contact layer, wherein the n-type contact includes a first noble metal; forming a p-type contact on the p-type contact layer, wherein the p-type contact includes a second noble metal; forming a conductive routing layer on the n-type contact and on the p-type contact, wherein the conductive routing layer is free of noble metals, wherein the conductive routing layer fully covers the n-type contact and the p-type contact; and forming a passivation layer over the conductive routing layer.

Inventors: **Yu; Chen-Hua (Hsinchu, TW), Lin; Yu-Hung (Taichung City, TW), Tsui; Ren-Fen (Taipei, TW)**

Applicant: **Taiwan Semiconductor Manufacturing Co., Ltd.** (Hsinchu, TW)

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application claims the benefit of U.S. Provisional Application No. 63/563,500, filed on Mar. 11, 2024, which application is hereby incorporated herein by reference.

BACKGROUND
[0002] Electrical signaling and processing are one technique for signal transmission and processing. Optical signaling and processing have been used in increasingly more applications in recent years, particularly due to the use of optical fiber-related applications for signal transmission.
[0003] Optical signaling and processing are typically combined with electrical signaling and processing to provide full-fledged applications. For example, optical fibers may be used for long-range signal transmission, and electrical signals may be used for short-range signal transmission as well as processing and controlling. Accordingly, devices integrating long-range optical components and short-range electrical components are formed for the conversion between optical signals and electrical signals, as well as the processing of optical signals and electrical signals. Packages thus may include both optical (photonic) dies including optical devices and electronic dies including electronic devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
[0005] FIGS. **1**, **2**, **3**, **4**, **5**, **6**, **7**, **8**, **9**, and **10** illustrate cross-sectional views of intermediate steps in the formation of a laser die, in accordance with some embodiments.
[0006] FIG. **11** illustrates a cross-sectional view of an intermediate step in the formation of a laser die, in accordance with some embodiments.
[0007] FIGS. **12**, **13**, **14**, **15**, and **16** illustrate cross-sectional views of intermediate steps in the formation of a laser die, in accordance with some embodiments.
[0008] FIGS. **17** and **18** illustrate cross-sectional views of intermediate steps in the formation of an interposer structure, in accordance with some embodiments.
[0009] FIGS. **19**, **20**, **21**, and **22** illustrate cross-sectional views of intermediate steps in the formation of a package, in accordance with some embodiments.
[0010] FIGS. **23**, **24**, **25**, and **26** illustrate cross-sectional views of intermediate steps in the formation of a package, in accordance with some embodiments.
[0011] FIGS. **27**, **28**, and **29** illustrate cross-sectional views of package structures, in accordance with some embodiments.
DETAILED DESCRIPTION
[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat

reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0013] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] Embodiments will now be discussed with respect to certain embodiments in which one or more laser dies and semiconductor dies are bonded to an optical interposer to form a package. The laser dies are manufactured with processes that allow for the utilization of noble metals (e.g., gold) or other materials that may be incompatible with other semiconductor fabrication processes. For example, the laser dies described herein utilize noble metals for electrical contacts and utilize more compatible metals for overlying electrical routing. In this manner, contamination from noble metals can be avoided while allowing for the benefits provided by the use of noble metals in electrical contacts. However, the embodiments presented herein are intended to be illustrative and are not intended to limit the embodiments to the precise descriptions as discussed. Rather, the embodiments discussed may be incorporated into a wide variety of implementations, and all such implementations are fully intended to be included within the scope of the embodiments.

[0015] FIGS. **1** through **16** illustrate intermediate steps in the formation of a laser die **50** (see FIG. **16**), in accordance with some embodiments. The laser die **50** is utilized to provide optical power (e.g., light) to optical components, photonic components, waveguides, or the like within a package. The laser die **50** may comprise light-generating structures such as the laser diode **34** described below for FIG. **9**. In particular embodiments the laser diode **34** may comprise a Fabry-Perot Diode, and may be based on III-V materials, II-VI materials, or any other suitable set of materials. The laser die **50** may have another configuration than shown, and may be formed using other materials or techniques than described for FIGS. **1-16**. For example, the embodiment described for FIGS. **1-16** includes an n-type first contact layer **12** with a corresponding n-type contact **24** (see FIG. **4**) and a p-type second contact layer **22** with a corresponding p-type contact **26** (see FIG. **5**), but in other embodiments, the first contact layer **12** may be p-type with a corresponding p-type contact and the second contact layer **22** may be n-type with a corresponding n-type contact. Other configurations are possible. In some embodiments, multiple laser dies **50** are formed on the same substrate **10** (see FIG. **1**) and/or the same support substrate **38** (see FIG. **12**) and then singulated into individual laser dies **50**.

[0016] FIG. **1** illustrates a stack **11** of material layers on a substrate **10**, in accordance with some embodiments. The stack **11** is subsequently patterned as part of forming the laser die **50**, in accordance with some embodiments. In some embodiments, the stack **11** comprises a first contact layer **12**, a first buffer layer **14**, an active layer **16** comprising multiple quantum wells (MQW), a second buffer layer **18**, a cladding layer **20**, and a second contact layer **22**. In other embodiments, other layers may be present in the stack **11**, or the layers of the stack **11** may have different characteristics than described herein. As an example, in other embodiments, an etch stop layer and/or a release layer may be present between the substrate **10** and the n-type contact layer **12**. The layers of the stack **11** may be formed using suitable techniques, such as using epitaxial growth processes or the like.

[0017] In some embodiments, the substrate **10** can provide structural support and act as a seed surface for epitaxially growing the overlying materials of the stack **11**. In some embodiments, the substrate **10** may be, for example, a 2-inch wafer, a 4-inch wafer, or the like. In particular embodiments in which the laser die **50** utilizes III-V materials to form the desired lasers, the substrate **10** may be a material such as indium phosphide, gallium arsenide, or gallium antimony. In other embodiments in which the laser die **50** utilizes II-VI materials to form the desired lasers, the substrate **10** may be a material such as gallium arsenide, cadmium telluride, zinc selenide, or the like. In still further embodiments, the substrate **10** may be a sapphire material, a semiconductor material, or a stack of multiple materials. All suitable materials may be utilized.

[0018] The first contact layer **12** is formed over the substrate **10**. The first contact layer **12** is subsequently patterned to form a first contact layer **12** of the laser diode **34**. For embodiments in which the laser diode **34** utilizes III-V compounds, the first contact layer **12** may comprise a material such as indium phosphide, gallium nitride, indium nitride, aluminum nitride, aluminum gallium nitride, aluminum indium nitride, aluminum indium gallium nitride, combinations thereof, or the like. Additionally, for embodiments in which the laser diode **34** utilizes II-VI compounds, the first contact layer **12** may use a III-V material such as indium phosphide, gallium arsenide, gallium antimonide, combinations of these, or the like. Other materials are possible.

[0019] In some embodiments, the first contact layer **12** may be doped with one or more dopants. For embodiments in which the first contact layer **12** is an n-type first contact layer **12**, the first contact layer **12** may be doped with one or more n-type dopants such as phosphorus, arsenic, antimony, bismuth, lithium, combinations thereof, or the like. In other embodiments in which the first contact layer **12** is a p-type first contact layer **12**, the first contact layer **12** may be doped with one or more p-type dopants such as boron, aluminum, gallium, indium, combinations thereof, or the like. However, any suitable dopants may be utilized. In some embodiments the first contact layer **12** is formed, for example, using an epitaxial growth process such as molecular beam epitaxy (MBE), hydride vapor phase epitaxy (HVPE), liquid phase epitaxy (LPE), or the like. The first contact layer **12** may be doped in situ during formation, although other processes may be utilized, such as ion implantation, diffusion, or the like.

[0020] In some embodiments, an optional etch stop layer or release layer (not pictured) may be formed between the substrate **10** and the first contact layer **12**. The etch stop layer or release layer may facilitate removal of the substrate **10** during subsequent processing steps (see FIG. **13**). The etch stop layer or release layer may comprise a suitable material, such as indium gallium arsenide, indium aluminum arsenide, doped indium phosphide, another material, a combination thereof, or the like.

[0021] The first buffer layer **14** is formed over the first contact layer **12** and is utilized in order to facilitate transition from the material of the first contact layer **12** to the material of the overlying layer (e.g., the active layer **16**), which can improve epitaxial growth of the overlying layers. In embodiments in which the laser diode **34** utilizes III-V compounds, the first buffer layer **14** may be a compound material such as indium gallium arsenide phosphide, indium gallium aluminum arsenide, indium gallium arsenide, combinations thereof, or the like. Additionally, in embodiments in which the laser diode **34** utilizes II-VI compounds, the first buffer layer **605** may be a II-VI material such as BeMgZnSe, BeZnCdSe, beryllium telluride, combinations of these, or the like. Additionally, the first buffer layer **14** may be deposited using an epitaxial growth process such as MBE, HVPE, LPE, or the like, and may be doped to the same type (e.g., n-type or p-type) as the first contact layer **12**. The first buffer layer **14** may be doped using similar dopants or techniques as the first contact layer **12**. However, any suitable material and any suitable method of deposition may be utilized.

[0022] The active layer **16** is formed over the first buffer layer **14**. The active layer **16** may be configured or designed to control the generation of light of desired wavelengths. For example, by adjusting and controlling the proportional composition of the elements in the active layer **16**, the bandgap of the materials in the active layer **16** may be adjusted, thereby adjusting the wavelength of light that will eventually be emitted.

[0023] In some embodiments, the active layer **16** comprises multiple quantum wells (MQW). MQW structures in the active layer **16** for embodiments which utilize III-V materials may comprise, for example, layers of indium aluminum gallium arsenide, indium gallium nitride, gallium nitride, aluminum indium gallium nitride, or the like. For embodiments which utilize II-VI based materials, the active layer **16** may comprise materials such as BeZnCdSe or the like. The active layer **16** may comprise any number of quantum wells, such as 5 to 20 quantum wells, for example, though other numbers of quantum wells are possible. In some embodiments, the active layer **16** may be epitaxially grown using metal organic chemical vapor deposition (MOCVD) with the first buffer layer **14** acting as a nucleation layer. Other processes, such as MBE, HVPE, LPE, or the like, may also be utilized.

[0024] The second buffer layer **18** is formed over the active layer **16** and is utilized in order to facilitate transition from the material of the active layer **16** to the material of the overlying layer (e.g., the cladding layer **20**), which can improve epitaxial growth of the overlying layers. In an embodiment in which the laser diode **34** utilizes III-V compounds, the second buffer layer **18** is a compound material such as indium gallium arsenide phosphide, indium gallium aluminum arsenide, indium gallium arsenide, combinations thereof, or the like. Additionally, in embodiments in which the laser diode **34** utilizes II-VI compounds, the second buffer layer **18** may be a II-VI material such as BeMgZnSe, BeZnCdSe, beryllium telluride, combinations of these, or the like. Additionally, the second buffer layer **18** may be deposited using an epitaxial growth process such as MBE, HVPE, LPE, or the like, and may be doped to the opposite type (e.g., n-type or p-type) as the first contact layer **12**. For example, the second buffer layer **18** may be doped as p-type when the first contact layer **12** is doped as n-type. However, any suitable material and any suitable method of deposition may be utilized.

[0025] The cladding layer **20** is formed over the second buffer layer **18**, and may facilitate transition from the material of the second buffer layer **18** to the material of the overlying layer (e.g., the second contact layer **22**), which can improve epitaxial growth of the overlying layers. The cladding layer **20** may facilitate optical confinement of the light generated in the active layer **16**, and may be considered a “ridge layer” in some cases. In an embodiment in which the laser diode **34** utilizes III-V compounds, the cladding layer **20** is a compound material such as indium phosphide or the like. Additionally, in embodiments in which the laser diode **34** utilizes II-VI compounds, the cladding layer **20** may be a II-VI material such as BeMgZnSe, BeZnCdSe, beryllium telluride, combinations of these, or the like. Additionally, the cladding layer **20** may be deposited using an epitaxial growth process such as MBE, HVPE, LPE, or the like, and may be doped to the opposite type (e.g., n-type or p-type) as the first contact layer **12**. For example, the cladding layer **20** may be doped as p-type when the first contact layer **12** is doped as n-type. However, any suitable material and any suitable method of deposition may be utilized.

[0026] The second contact layer **22** is formed over the cladding layer **20**. The second contact layer **22** is subsequently patterned to form a second contact layer **22** of the laser diode **34**. In an embodiment in which the laser diode **34** is based on III-V materials, the second contact layer **22** may comprise a group III-V compound material such as indium gallium arsenide, indium aluminum arsenide, gallium nitride, indium nitride, aluminum nitride, aluminum gallium nitride, aluminum indium nitride, aluminum indium gallium nitride, combinations thereof, or the like. Additionally, the second contact layer **22** may be deposited using an epitaxial growth process such as MBE, HVPE, LPE, or the like, and may be doped to the opposite type (e.g., n-type or p-type) as the first contact layer **12**. For example, the cladding layer **20** may be doped as p-type when the first contact layer **12** is doped as n-type. In some embodiments, the second contact layer **22** may be doped with one or more p-type dopants such as boron, aluminum, gallium, indium, combinations thereof, or the like. However, any suitable material and any suitable methods of deposition or doping may be utilized.

[0027] In FIG. **2**, the stack **11** is patterned to form a laser diode structure, in accordance with some embodiments. The stack **11** may be patterned using suitable photolithographic masking and etching processes. Each etching process may include one or more wet etches and/or dry etches. In some embodiments, the second contact layer **22** and the cladding layer **20** are patterned using a first photolithographic masking and etching process. The second buffer layer **18**, the active layer **16**, and the first buffer layer **14** are patterned using a second photolithographic masking and etching process. The first contact layer **12** is patterned using a third photolithographic masking and etching process. For example, the first contact layer **12** is patterned to become a first contact layer **12**. In some embodiments, the first contact layer **12** may have an adiabatic taper or other shape that facilitates evanescent optical coupling to underlying layers, structures, or waveguides. However, any suitable patterning processes and/or any suitable number of patterning processes may be utilized to pattern the stack **11** in other embodiments.

[0028] In FIG. **3**, a recess **23** is formed in a top surface of the first contact layer **12**, in accordance with some embodiments. An n-type contact **24** (see FIG. **4**) is subsequently formed in the recess **23**. The recessed surfaces of the first contact layer **12** within the recess **23** may also be referred to as the "contact area **23**" of the first contact layer **12**. In this manner, the contact area **23** of the first contact layer **12** is recessed from the top surface of the first contact layer **12**. Forming a recess **23** as described herein may allow for improved physical isolation and reduced contact resistance of the subsequently formed n-type contact **24**. The recess **23** may be formed using a suitable photolithographic masking and etching process. For example, a mask (e.g. a photoresist, hardmask, or the like) may be formed over the structure and patterned using photolithographic techniques, in which the pattern corresponds to the recess **23**. The first contact layer **12** may then be etched using the patterned mask as an etch mask to form the recess **23**. The etching may include one or more wet etching processes and/or dry etching processes. In some embodiments, the patterned mask is removed after forming the recess **23**, and in other embodiments, the patterned mask is left remaining for use during formation of the n-contact **24**, described below. In some embodiments, the recess **23** has a depth (e.g., from a top surface of the first contact layer **12**) that is in the range of about 0.01 μm to about 0.5 μm. In some embodiments, the recess **23** has a length or width that is in the range of about 50 μm to about 2000 μm. Other dimensions are possible.

[0029] In FIG. **4**, one or more conductive layers **25** are deposited in the recess **23** to form the n-type contact **24**, in accordance with some embodiments. For example, the n-type contact **24** shown in FIG. **4** includes three conductive layers **25**, indicated as bottom conductive layer **25**A, middle conductive layer **25**B, and top conductive layer **25**C. In other embodiments, an n-type contact **24** may have only one conductive layer **25**, two conductive layers **25**, or more than three conductive layers **25**. An example of an n-type contact **24** comprising four conductive layers **25**A-D is described below for FIG. **11**. The conductive layers **25** may comprise one or more conductive materials such as gold, germanium, nickel, titanium, other metals, alloys thereof, combinations thereof, or the like. The conductive layers **25** of an n-type contact **24** may be formed of different materials, the arrangement of which may be chosen to provide desired properties. For example, with reference to FIG. **4**, the bottom conductive layer **25**A may be gold, the middle conductive layer **25**B may be germanium, and the top conductive layer **25**C may be nickel. This is an example, and other materials or combinations of materials are possible. The conductive layers **25** of the n-type contact **24** may be deposited using one or more suitable deposition techniques, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), plating, or the like.

[0030] In some embodiments, a mask may be formed over the laser diode structure and patterned (e.g., using photolithographic techniques) to expose the contact area **23**. The materials of the conductive layers **25** may be deposited over the patterned mask and over the contact area **23**. The patterned mask may then be removed, leaving the n-type contact **24** formed on the contact area **23**. In other embodiments, the patterned mask used to form the recess **23** may be left remaining over the structure and used during deposition of the conductive layers **25**. In some embodiments, the n-type contact **24** is formed only within or over the recess **23**. In other words, the n-type contact **24** is formed only on the contact area **23** of the first contact layer **12**. Accordingly, the entire n-type contact **24** is inside the perimeter of the first contact layer **12**, and the entire n-type contact **24** laterally overlaps the first contact layer **12**. In some embodiments, the n-type contact **24** does not extend on or over the top surface of the first contact layer **12**. In some embodiments, entire bottom surface of the n-type contact **24** physically contacts a surface of the first contact layer **12**. Forming the n-type contact **24** in the recess **23** (e.g., only on the contact area **23**) may allow for improved contact quality and improved physical isolation that can reduce contamination, described in greater detail below.

[0031] In some embodiments, each conductive layer **25** of the n-type contact **24** may have a thickness in the range of about 10 nm to about 500 nm, and the n-type contact **24** may have an overall thickness in the range of about 30 nm to about 1.5 μm. Other thicknesses are possible. A thickness of a conductive layer **25** or the overall thickness of the n-type contact **24** may be greater than, less than, or about the same as the depth of the recess **23**. The top surface of the n-type contact **24** may be lower than, approximately level with, or protrude from the top surface of the first contact layer **12**. For example, FIG. **4** shows the n-type contact **24** protruding from the top surface of the first contact layer **12**.

[0032] In some embodiments, one or more conductive layers **25** of the n-type contact **24** may be fully or partially within the recess **23**, and may or may not have exposed surfaces. For example, the bottom conductive layer **25**A shown in FIG. **4** is fully within the recess **23** such that surfaces of the bottom conductive layer **25**A are fully covered by the first contact layer **12** and the middle conductive layer **24**B. In other embodiments, more than one conductive layer **25** may be fully within the recess **23**. In some cases, fully covering a conductive layer **25** can reduce the chance of contamination. For example, covering a conductive layer **25** may reduce contamination by the material of the conductive layer **25** during subsequent processing. In some cases, materials such as noble metals can cause process defects or material defects during processing, and covering such a conductive layer **25** material can reduce the chance of defects due to contamination.

[0033] In FIG. **5**, one or more conductive layers **27** are deposited on the second contact layer **22** to form the p-type contact **26**, in accordance with some embodiments. For example, the p-type contact **26** shown in FIG. **5** includes two conductive layers **27**, indicated as bottom conductive layer **27**A and top conductive layer **27**B. In other embodiments, a p-type contact **26** may have only one conductive layer **27** or more than two conductive layers **27**. An example of a p-type contact **26** comprising three conductive layers **27**A-C is described below for FIG. **11**. The conductive layers **27** may comprise one or more conductive materials such as platinum, titanium, nickel, palladium, silver, gold, other metals, alloys thereof, combinations thereof, or the like. The conductive layers **27** of a p-type contact **26** may be formed of different materials, the arrangement of which may be chosen to provide desired properties. For example, with reference to FIG. **5**, the bottom conductive layer **27**A may be titanium and the top conductive layer **27**B may be platinum. This is an example, and other materials or combinations of materials are possible. The conductive layers **27** of the p-type contact **26** may be deposited using one or more suitable deposition techniques, such as CVD, PVD, ALD, plating, or the like.

[0034] In some embodiments, a mask may be formed over the structure and patterned (e.g., using photolithographic techniques) to expose the second contact layer **22**. The materials of the conductive layers **27** may be deposited over the patterned mask and over the second contact layer **22**. The patterned mask may then be removed, leaving the p-type contact **26** formed on the second contact layer **22**. In some embodiments, the p-type contact **26** is formed only on the second contact layer **22**. As shown in FIG. **5**, a width of the p-type contact **26** may be smaller than a width of the second contact layer **22**. Accordingly, the entire p-type contact **26** is inside the perimeter of the second contact layer **22**, and the entire p-type contact **26** laterally overlaps the second contact layer **22**. In some embodiments, entire bottom surface of the p-type contact **26** physically contacts a surface of the second contact layer **22**. In some embodiments, each conductive layer **27** of p-type contact **26** may have a thickness in the range of about 10 nm to about 500 nm, and the p-type contact **26** may have an overall thickness in the range of about 30 nm to about 1.5 μm. Other thicknesses are possible.

[0035] In FIG. **6**, an insulating layer **28** is formed over the structure, in accordance with some embodiments. The insulating layer **28** protects and electrically isolates the first contact layer **12**, the n-type contact **24**, the first buffer layer **14**, the active layer **16**, the second buffer layer **18**, the cladding layer **20**, the second contact layer **22**, and the p-type contact **26**. Accordingly, the insulating layer **28** may comprise one or more insulating materials such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof. The insulating layer **28** may be deposited using a suitable technique such as CVD, ALD, PVD, the like, or a combination thereof. In some embodiments, the insulating layer **28** may be conformally deposited in a blanket layer. However, any suitable materials and any suitable methods of deposition may be utilized. In some embodiments, the insulating layer **28** has a thickness in the range of about 0.1 μm to about 5 μm, though other thicknesses are possible.

[0036] In FIG. **7**, the insulating layer **28** is patterned to expose the n-type contact **24** and the p-type contact **26**, in accordance with some embodiments. The patterning of the insulating layer **28** may form openings in the insulating layer **28** that expose the n-type contact **24** and the p-type contact **26**. In other cases, the patterning of the insulating layer **28** may remove portions of the insulating layer **28** overlying the n-type contact **24** and the p-type contact **26**. In some embodiments, the patterning exposes top surfaces of the n-type contact **24** and the p-type contact **26**. In an embodiment the patterning may be performed using, e.g., a photolithographic masking and etching process. However, any suitable patterning process may be utilized. In some embodiments, top surfaces of the n-type contact **24** and/or the p-type contact **26** may be approximately level with corresponding top surfaces of the insulating layer **28**. In other embodiments, top surfaces of the n-type contact **24** and/or the p-type contact **26** may be recessed from or protrude from corresponding top surfaces of the insulating layer **28**.

[0037] In other embodiments, the insulating layer **28** is deposited before forming the n-type contact **24** and the p-type contact **26**. One or more openings in the insulating layer **28** may then be patterned to expose the contact areas of the first contact layer **12** and the second contact layer **22**. The conductive layers **25** of the n-type contact **24** and the conductive layers **27** of the p-type contact **26** may then be deposited through the opening(s) in the insulating layer **28**. In some embodiments, a mask (e.g., a photoresist) may be deposited over the insulating layer **28** and patterned to expose the opening(s) in the insulating layer **28**, thus exposing the first contact layer **12** and/or the second contact layer **22**. The conductive layers **25** and/or the conductive layers **27** may be deposited over the mask and in the opening(s). The mask may then be removed, with the remaining portions of the conductive layers **25** and/or the conductive layers **27** forming the n-type contact **24** and/or the p-type contact **26**. Separate patterned masks may be used to form the n-type contact **24** and the p-type contact **26**, in some embodiments. This is an example, and other processes to form the n-type contact **24** and the p-type contact **26** are possible.

[0038] In FIG. **8**, an adhesion layer **30** is deposited over the exposed surfaces of the n-type contact **24** and the p-type contact **26**, in accordance with some embodiments. The adhesion layer **30** may facilitate adhesion of the overlying routing layers **32** (see FIG. **9**) and protect the contacts **24**/**26**, in accordance with some embodiments. The adhesion layer **30** makes physical and electrical contact to the n-type contact **24** and the p-type contact **26**. For example, one region of the adhesion layer **30** is formed on the n-type contact **24** and another region of the adhesion layer **30** is formed on the p-type contact **26**. As shown in FIG. **8**, the adhesion layer **30** may also extend on surfaces of the insulating layer **28**. The adhesion layer **30** may comprise one or more suitable conductive materials such as titanium, titanium nitride, tantalum, tantalum nitride, the like, or a combination thereof. The adhesion layer **30** may be deposited using a suitable technique, such as CVD, PVD, ALD, the like, or a combination thereof. However, any suitable materials and any suitable methods of deposition may be utilized.

[0039] In FIG. **9**, the routing layers **32** are deposited on the adhesion layer **30** to form the laser diode **34**, in accordance with some embodiments. The laser diode **34** comprises the layers and structures formed on the substrate **10**, as shown in FIG. **9**. The routing layers **32** provide electrical routing for the laser diode **34** and allow for external electrical connections to be made to the laser diode **34**. In some embodiments, the routing layers **32** comprise one or more layers of conductive materials such as aluminum, copper, aluminum copper alloy, another metal, the like, or a combination thereof. In some embodiments, the routing layers **32** are free of materials that can cause undesirable contamination during processing. For example, the routing layers **32** may be free of gold, other noble metals, or the like that can cause contamination or process defects. In some embodiments, the materials such as gold, other noble metals, or the like may be present only in the n-type contact **24** and/or the p-type contact **26** and may not be present in the routing layers **32**. In this manner, contamination and process defects can be reduced while still allowing for quality electrical routing and/or electrical contacts to be formed.
[0040] The routing layers **32** may be formed using a suitable deposition process, such as CVD, PVD, ALD, plating, or a combination thereof. However, any suitable material and method of manufacture may be utilized. In an embodiment in which the routing layers **32** are plated, the routing layers **32** may be patterned during the deposition process. In other embodiments, the routing layers **32** may be patterned after deposition using, for example, a photolithographic masking and etching process. However, any suitable process may be utilized. In some embodiments, the routing layers **32** may have a thickness in the range of about 0.3 μm to about 5 μm, though other thicknesses are possible. In other embodiments, an optional additional adhesion layer (not shown) may be formed on the routing layers **32**. In embodiments in which the additional adhesion layer is formed, the additional adhesion layer may be similar to the adhesion layer **30** and may be formed using similar materials or techniques.
[0041] In FIG. **10**, a passivation layer **36** is formed over the laser diode **34**, in accordance with some embodiments. In an embodiment, the passivation layer **36** is a protective dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, combinations of these, or the like. The passivation layer **36** may be deposited using a deposition process such as CVD, PVD, ALD, combinations of these, or the like. However, any suitable materials and methods may be used to form the passivation layer **36**. In some embodiments, a planarization process may be performed to planarize the top surface of the passivation layer **36**. The planarization process may comprise a chemical mechanical polish (CMP) process, a grinding process, or the like.
[0042] FIG. **11** illustrates a laser diode **34**, in accordance with some embodiments. The structure shown in FIG. **11** is similar to the structure shown in FIG. **10**, except that the n-type contact **24** and the p-type contact **26** comprise another number of conductive layers **25** and conductive layers **27**, respectively. For example, the n-type contact **24** of FIG. **11** comprises four conductive layers **25**, indicated as conductive layers **25**A, **25**B, **25**C, and **25**D, and the p-type contact **26** of FIG. **11** comprises three conductive layers **27**, indicated as conductive layers **27**A, **27**B, and **27**C. In some embodiments, the topmost conductive layer **25**D of the n-type contact **24** and/or the topmost conductive layer **27**C of the p-type contact **26** may comprise gold or the like. In some cases, forming a topmost conductive layer **25**D/**27**C of gold can improve the conductive properties of the n-type contact **24** and the p-type contact **26**. In some embodiments, a topmost conductive layer **25**D/**27**C of gold is covered by the adhesion layer **30** and the routing layers **32** to reduce the chance of contamination.
[0043] FIGS. **12** through **16** illustrate additional intermediate steps in the formation of a laser die **50**, in accordance with some embodiments. The process described for FIGS. **12-16** is shown using the laser diode **34** described for FIG. **10**, but all other embodiments (e.g., the laser diode **34** of FIG. **11**) described herein and variations thereof may be used. Accordingly, the process shown in FIGS. **12-16** follows from FIG. **10**, in some embodiments.
[0044] In FIG. **12**, the structure is flipped over and bonded to a support substrate **38**, in accordance

with some embodiments. For example, the passivation layer **36** may be bonded to the support substrate **38** using fusion bonding, direct bonding, dielectric-to-dielectric bonding, or the like. In some embodiments, the support substrate **38** may comprise silicon (e.g., a silicon wafer), a metal, a ceramic, or the like. In some embodiments, the bonding process may comprise performing an activation process on bonding surfaces of the passivation layer **36** and the support substrate **38** and then placing the passivation layer **36** and the support substrate **38** in physical contact to initiate the bonding process. A thermal process or the like may be used to strengthen the bond, in some cases. However, any other suitable attachment process, including using an adhesive, may be utilized.
[0045] In FIG. **13**, the substrate **10** is removed, in accordance with some embodiments. After removal of the substrate **10**, surfaces of the first contact layer **12** and/or the insulating layer **28** may be exposed. The substrate **10** may be removed using a CMP process, a grinding process, the like, or a combination thereof. In other embodiments, the substrate **10** may be removed using one or more etching processes, which may include wet etching processes and/or dry etching processes. The etching processes may be selective to the material of the substrate **10**, and may stop on an etch stop layer (not shown) between the substrate **10** and the laser diode **34**. In some embodiments, a planarization process may be performed after removal of the substrate **10**. Other removal techniques are possible.
[0046] In FIG. **14**, a dielectric layer **40** is deposited over the laser diode **34**, in accordance with some embodiments. The dielectric layer **40** may comprise one or more layers of dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, the like, or a combination thereof. In some embodiments, the dielectric layer **40** comprises a dielectric material suitable for dielectric-to-dielectric bonding. The dielectric layer **40** may be deposited using one or more processes, such as CVD, PVD, ALD, spin-on, or the like. Any suitable materials, deposition techniques, or number of layers may be used.
[0047] In FIG. **15**, openings **41** are formed through the dielectric layer **40** and the insulating layer **28**, in accordance with some embodiments. The openings **41** may extend through the dielectric layer **40** and the insulating layer **28** to expose the adhesion layer **30**. In some embodiments, the openings **41** may extend through the adhesion layer **30** to expose the routing layers **32**. The openings **41** may be formed using suitable photolithographic masking and etching techniques.
[0048] In FIG. **16**, bonding pads **42** are formed in the openings **41** to form a laser die **50**, in accordance with some embodiments. The bonding pads **42** are conductive features that make electrical contact to the routing layers **32** and allow for electrical connection between the laser die **50** and external structures. For example, at least one bonding pad **42** may be electrically connected to the n-type contact **24** and at least one bonding pad **42** may be electrically connected to the p-type contact **26**. In some cases, the bonding pads **42** may comprise via portions that physically contact the n-type contact **24** and the p-type contact **26**.
[0049] In some embodiments, the bonding pads **42** may be formed by depositing a conductive material in the openings **41**. In an embodiment, the conductive material may comprise a barrier layer, a seed layer, a fill metal, or a combination thereof. For example, a barrier layer may first be blanket deposited over the dielectric layer **40** and within the openings **41**. The barrier layer may comprise titanium, titanium nitride, tantalum, tantalum nitride, the like, or a combination thereof. The seed layer may be a conductive material such as copper and may be blanket deposited over the barrier layer using a suitable process, such as sputtering, evaporation, plasma-enhanced chemical vapor deposition (PECVD), or the like. The fill metal may be a conductive material such as copper, copper alloy, aluminum, or the like, and may be deposited using a suitable process, such as electroplating, electroless plating, or the like. The fill metal may fill or overfill the openings **41**, in some embodiments. Once the fill metal has been deposited, excess material of the fill metal, the seed layer, and the barrier layer may be removed using, for example, a planarization process such as a CMP process or the like. After the planarization process, top surfaces of the dielectric layer **40** and the bonding pads **42** may be substantially level or coplanar, in some cases. Other materials or

techniques are possible.

[0050] FIGS. **17** through **22** illustrate intermediate steps in the formation of a package **100**, in accordance with some embodiments. In some embodiments, the package **100** comprises a laser die **50** and a semiconductor die **150** bonded to an interposer structure **101**. In some embodiments, the laser die **50** provides optical power to waveguides **112** within the interposer structure **101**. The package **100** described for FIGS. **17-22** is an example, and other configurations or arrangements are possible. For example, in other embodiments, more than one laser die **50** or more than one semiconductor die **150** may be bonded to the interposer structure **101**. The interposer structure **101** or semiconductor die **150** may have other configurations, arrangements, or features than shown. All such variations are considered within the scope of the present disclosure. In some cases, the interposer structure **101** may be considered to be an optical interposer. In some cases, the package **100** may be considered to be an integrated chip, a photonic chip, an optical die, or the like.

[0051] FIGS. **17** and **18** illustrate intermediate steps in the formation of an interposer structure **101**, in accordance with some embodiments. The interposer structure **101** comprises interconnect layers **106** on a substrate **102**, in accordance with some embodiments. The substrate **102** may be a wafer, such as a silicon wafer, in some embodiments. Other substrates, such as a silicon-on-insulator (SOI) substrate, a multi-layered substrate, or a gradient substrate may also be used. The substrate **102** may be doped (e.g., with a p-type or an n-type dopant) or undoped. In some embodiments, the semiconductor material of the substrate **102** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. In other embodiments, the substrate **102** may be a dielectric material such as silicon oxide, glass, ceramic, plastic, polyimide, or any other suitable material that allows for structural support of overlying devices. In some embodiments, multiple interposer structures **101** may be formed on a single substrate **102** and then may be subsequently singulated into individual interposer structures **101** or individual packages **100**. In some embodiments, active devices (e.g., transistors, diodes, or the like), passive devices (e.g. capacitors, resistors, or the like), integrated circuits, and/or the like may be formed in the substrate **102**. The substrate **102** may be free of passive or active devices, in other embodiments.

[0052] In some embodiments, the interposer structure **101** comprises through vias **104** extending into the substrate **102**. The through vias **104** are electrically connected to the interconnect layers **106**. The through vias **104** may be formed, for example, by forming openings extending into the substrate **102**. The openings may be formed using acceptable photolithography and etching techniques, such as by forming and patterning a photoresist and then performing an etching process using the patterned photoresist as an etching mask. The etching process may include, for example, a dry etching process and/or a wet etching process. A conductive material may then be formed in the openings, thereby forming the through vias **104**. In some embodiments, a liner (not shown) may be deposited in the openings prior to forming the conductive material. The conductive material may comprise, for example, a metal or a metal alloy such as copper, silver, gold, tungsten, cobalt, aluminum, alloys thereof, or the like. A planarization process (e.g., a CMP process or a grinding process) may be performed to remove excess conductive material along the surface of the substrate **102** such that surfaces of the through vias **104** and the substrate **102** are level. The through vias **104** may protrude from the substrate **102** and into the interconnect layers **106**, in other embodiments. Other materials or techniques are possible.

[0053] The interconnect layers **106** comprise one or more layers of conductive features **105** formed in one or more dielectric layers **107** (not individually illustrated), in some embodiments. The conductive features **105** may comprise conductive lines, conductive vias, conductive pads, metallization patterns, redistribution layers, or the like that provide electrical interconnections and electrical routing. In some embodiments, the interconnect layers **106** may have multiple layers of conductive features **105**, but the precise number of layers of conductive features **105** may be dependent upon the design of the interposer structure **101**. The conductive features **105** may be formed using any suitable techniques such as deposition, damascene, dual damascene, or the like. The conductive features **105** may include, for example, a metal or a metal alloy such as copper, silver, gold, tungsten, cobalt, ruthenium, aluminum, alloys thereof, combinations thereof, or the like. Other materials are possible.

[0054] Acceptable dielectric materials for the dielectric layers **107** include oxides such as silicon oxide or aluminum oxide; nitrides such as silicon nitride; carbides such as silicon carbide; the like; or combinations thereof such as silicon oxynitride, silicon oxycarbide, silicon carbonitride, silicon oxycarbonitride or the like. Other dielectric materials may also be used, such as a polymer such as polybenzoxazole (PBO), polyimide, a benzocyclobuten (BCB) based polymer, or the like. The dielectric layers **107** may be formed using any suitable techniques. In some embodiments, the interconnect layers **106** may have multiple dielectric layers **107**, but the precise number of dielectric layers **107** may be dependent upon the design of the interposer structure **101**.

[0055] In FIG. **18**, interconnect layers **108** are formed over the interconnect layers **106** to form the interposer structure **101**, in accordance with some embodiments. The interconnect layers **106** and the interconnect layers **108** may collectively considered to be an interconnect structure **110**, in some cases. In some embodiments, the interconnect layers **108** comprise bonding pads **111** and waveguides **112** formed in one or more dielectric layers **109** (not individually illustrated). The interconnect layers **108** may also comprise conductive features similar to the conductive features **105**, in some cases. The dielectric layers **109** may be similar to the dielectric layers **107**, in some cases. In some embodiments, the topmost dielectric layer **109** is a bonding layer formed of a material suitable for dielectric-to-dielectric bonding.

[0056] In some embodiments, the bonding pads **111** are formed in the topmost dielectric layer **109** of the dielectric layers **109**. The bonding pads **111** are electrically coupled to underlying conductive features, which may include conductive features **105**. In some cases, the bonding pads **111** may also be considered conductive features of the interconnect layers **108**. The bonding pads **111** may be formed using suitable techniques, such as deposition, damascene, dual damascene, or the like. The bonding pads **111** may include, for example, a metal or a metal alloy such as copper, aluminum, alloys thereof, combinations thereof, or the like. Other materials are possible. In some embodiments, a planarization process (e.g., a CMP process or grinding process) may be performed to level surfaces of the topmost dielectric layer **109** and the bonding pads **111**.

[0057] The waveguides **112** may allow for the transmission of optical power and/or optical signals within the interposer structure **101**. FIG. **18** shows two layers of waveguides **112** formed within the dielectric layers **109**, but another number of layers of waveguides **112** may be formed in other embodiments. In some embodiments, a waveguide **112** may be optically coupled to an adjacent waveguide **112**, to an overlying waveguide **112** of another layer, and/or to an underlying waveguide **112** of another layer. Waveguides **112** may be optically coupled using suitable techniques, such as using evanescent coupling, grating couplers, or other optical coupling techniques. The waveguides **112** may have a different configuration or arrangement than shown.

[0058] In some embodiments, a layer of waveguides **112** may be formed by depositing a waveguide material on a dielectric layer **109** and then patterning the waveguide material. In some embodiments, the waveguide material may be deposited on the dielectric layer **109** and thus the resulting waveguides **112** are formed on the dielectric layer **109**. In other cases, the waveguide material is deposited on a previously deposited dielectric layer **109**. The waveguide material may be a dielectric material such as silicon nitride, silicon oxide, silicon oxynitride, polymer, combinations of these, or the like. In other embodiments, the waveguide material may be a semiconductor material such as silicon, germanium, or the like. The waveguide material may be deposited using a suitable technique, such as CVD, PVD, ALD, or the like. The waveguide material may then be patterned using suitable photolithographic masking and etching techniques to form a layer of waveguides **112**. Another dielectric layer **109** may then be deposited over the waveguides **112**. The steps of depositing a waveguide material, patterning the waveguide material to form a layer of waveguides **112**, and then depositing a dielectric layer **109** over the layer of waveguides **112** may be repeated to form multiple layers of waveguides **112** within the interconnect layers **108**.

[0059] FIGS. **19** and **20** illustrate the bonding of a laser die **50** and a semiconductor die **150** to the interposer structure **101**, in accordance with some embodiments. FIG. **19** illustrates the laser die **50** and semiconductor die **150** prior to bonding to the interposer structure **101**, and FIG. **20** illustrates the laser die **50** and semiconductor die **150** after bonding to the interposer structure **101**. The laser die **50** may be similar to the laser die **50** described previously for FIG. **16**. In some embodiments, the laser die **50** and/or the semiconductor die **150** may have a thickness in the range of about 50 μm to about 700 μm, though other thicknesses are possible. The semiconductor die **150** shown and described is an illustrative example, and other dies, packages, components, or the like may be used in other embodiments.

[0060] The semiconductor die **150** may be a die, a chip, a system-on-chip (SoC) device, a system-on-integrated-circuit (SoIC) device, a package, a component, the like, or a combination thereof. In some cases, the semiconductor die **150** comprises an electronic die **160** connected to an interconnect structure **152**. A support structure **162** may be over the electronic die **160** and may be similar to the support substrate **38** described previously, in some embodiments. For example, in some cases, the support structure **162** may be silicon or the like. The interconnect structure **152** may comprise a plurality of conductive features, waveguides **156**, photonic components **158**, and/or other features formed in dielectric layers. In some cases, the interconnect structure **152** may be considered a photonic integrated circuit (PIC) or the like. The interconnect structure **152** may include multiple layers of conductive features formed in multiple dielectric layers. The conductive features may include conductive lines, conductive vias, conductive pads, or the like, and may include bonding pads **154** formed in a bonding layer. The interconnect structure **152** may or may not include active devices.

[0061] In some embodiments, the photonic components **158** may include such devices or components as optical waveguides (e.g., ridge waveguides, rib waveguides, buried channel waveguides, diffused waveguides, etc.), couplers (e.g., grating couplers, edge couplers comprising a tip waveguide having a width in the range of about 1 nm to about 200 nm, etc.), directional couplers, optical modulators (e.g., germanium modulators, Mach-Zehnder silicon-photonic switches, microelectromechanical switches, micro-ring resonators, etc.), amplifiers, multiplexors, demultiplexors, optical-to-electrical converters (e.g., photodetectors, P-N junctions, or the like), electrical-to-optical converters, lasers (e.g., laser diodes), phase shifters, combinations of these, or the like. However, the photonic components **158** may comprise other devices, structures, or components than these examples. The photonic components **158** may be optically coupled to waveguides **156** of the interconnect structure **152**. The waveguides **156** may be formed of a suitable material such as silicon, germanium, silicon nitride, a polymer, or the like.

[0062] In some embodiments, an electronic die **160** is connected to the interconnect structure **152** such that the electronic die **160** is electrically coupled to the interconnect structure **152**. In other embodiments, multiple electronic dies **160** are connected to the interconnect structure **152**, or an electronic die **160** may have other dimensions than shown. The electronic die **160** may be bonded to the interconnect structure **152** using, for example, fusion bonding or solder bumps.

[0063] The electronic die **160** may comprise, for example, a chip, a die, a system-on-chip (SoC) device, a system-on-integrated-circuit (SoIC) device, the like, or a combination thereof. For example, the electronic die **160** may include controllers, drivers, transimpedance amplifiers, transistors, other active devices, resistors, capacitors, other passive devices, the like, or combinations thereof. Accordingly, the electronic die **160** may be considered an electronic integrated circuit (EIC) or the like. In some embodiments, the integrated circuits may be configured to interface with the photonic components **158**. For example, the integrated circuits may be configured to control the operation of the photonic components **158**, to process electronic signals received from the photonic components **158**, or the like. The integrated circuits may be configured to control high-frequency signaling of the photonic components **158** according to received electrical signals (digital or analog). In this manner, the electronic die **160** may process or transmit electrical signals based on received optical signals and/or may process or transmit optical signals based on received electrical signals.

[0064] In some embodiments, the electronic die **160** may provide Serializer/Deserializer (SerDes) functionality. In some embodiments, the electronic die **160** may comprise one or more processing devices, such as a Central Processing Unit (CPU or "xPU"), a Graphics Processing Unit (GPU), an Application-Specific Integrated Circuit (ASIC), a High-Performance Computing (HPC) die, a Micro Control Unit (MCU) die, a BaseBand (BB) die, an Application processor (AP) die, an Application-Specific Integrated Circuit (ASIC) die, a logic die, the like, or a combination thereof. The electronic die **160** may include one or more memory devices, which may be a volatile memory such as Dynamic Random-Access Memory (DRAM), Static Random-Access Memory (SRAM), High-Bandwidth Memory (HBM), another type of memory, or the like. Other types or configurations of electronic dies **160** are possible.

[0065] In FIG. **20**, the laser die **50** and the semiconductor die **150** are bonded to the interposer structure **101**, in accordance with some embodiments. For example, the bonding pads **42** of the laser die **50** and the bonding pads **154** of the semiconductor die **150** may be bonded to corresponding bonding pads **111** of the interconnect structure **110** using metal-to-metal bonding (e.g., fusion bonding, direct bonding, or the like). In some embodiments, bonding layers of the laser die **50** and/or the semiconductor die **150** are bonded to a bonding layer of the interconnect structure **110** using dielectric-to-dielectric bonding (e.g., oxide-to-oxide bonding, fusion bonding, direct bonding, or the like).

[0066] In some embodiments, the bonding process may be initiated by activating the bonding surfaces and/or bonding pads of the laser die **50**, the semiconductor die **150**, and/or the interconnect structure **110**. Activating the bonding surfaces may comprise, for example, a dry treatment, a wet treatment, a plasma treatment, exposure to an inert gas plasma, exposure to H.sub.2, exposure to N.sub.2, exposure to O.sub.2, combinations thereof, or the like. For embodiments in which a wet treatment is used, an RCA cleaning process may be used, for example. In other embodiments, the activation process may comprise other types of treatments. After the activation process, the laser die **50** and the semiconductor die **150** are aligned and placed into physical contact with the interconnect structure **110**. The laser die **50**, the semiconductor die **150**, and the interposer structure **101** are then subjected to a thermal treatment and contact pressure to bond respective bonding layers together with dielectric-to-dielectric bonding and bond the corresponding bonding pads together with metal-to-metal bonding. In some embodiments, the resulting bonded structure is subsequently baked, annealed, pressed, or otherwise treated to strengthen or finalize the bond. This is an example, and other bonding processes are possible.

[0067] In some embodiments, after bonding, waveguides **112** of the interconnect structure **110** are optically coupled to the laser diode **34** of the laser die **50** and/or to waveguide(s) **156** of the semiconductor die **150**. In this manner, optical power may be transmitted from the laser die **50** to waveguide(s) **112**, and then may be transmitted from waveguide(s) **112** to the semiconductor die **150**. In some embodiments, optical signals may be transmitted between waveguide(s) **112** of the interconnect structure **110** and waveguide(s) **156** of the semiconductor die **150**. In this manner, optical communication within a package or between components thereof may be achieved.

[0068] In FIG. **21**, an encapsulant **116** is formed on and around the laser die **50** and the semiconductor die **150**, in accordance with some embodiments. After formation, the encapsulant **116** encapsulates the laser die **50** and the semiconductor die **150**. The encapsulant **116** may be a molding compound, an epoxy, a polymer, a composite material, a dielectric material, or the like. In some embodiments, the encapsulant **116** is applied by deposition, spin-on, compression molding, transfer molding, or the like. The encapsulant **116** may be formed over the interconnect structure **110** such that the laser die **50** and the semiconductor die **150** are buried or covered. The encapsulant **116** may be applied in liquid or semi-liquid form and then subsequently cured.

[0069] In FIG. **22**, a planarization process is performed on the encapsulant **116** to expose the laser die **50** and the semiconductor die **150**, in accordance with some embodiments. Top surfaces of the laser die **50**, the semiconductor die **150**, and the encapsulant **116** may be substantially level or coplanar (within process variations) after performing the planarization process. The planarization process may comprise, for example, a chemical-mechanical polish (CMP) process, a grinding process, an etching process, or the like. In other embodiments, a planarization process is not performed.

[0070] Further in FIG. **22**, interconnect layers **120** are formed on the back side of the substrate **102** as part of the interposer structure **101**, in accordance with some embodiments. In some embodiments, material is first removed from the back side of the substrate **102** to expose the through vias **104**. The material of the substrate **102** may be removed using a CMP process, a grinding process, an etching process, the like, or a combination thereof. After removing the material of the substrate **102**, surfaces of the through vias **104** and the substrate **102** may be substantially level or coplanar. After exposing the through vias **104**, interconnect layers **120** are formed on the through vias **104** and on the back side of the substrate **102**, in accordance with some embodiments. The interconnect layers **120** may be similar to the interconnect layers **106**, and may be formed using similar materials or techniques. For example, the interconnect layers **120** may comprise one or more layers of conductive features formed in one or more dielectric layers.

[0071] In some embodiments, conductive connectors **130** are formed on the back side of the interposer structure **101** to form the package **100**, in accordance with some embodiments. The conductive connectors **130** are physically and electrically connected to conductive features of the interconnect layers **120**, and may be used to connect the package **100** to an external component or package substrate. The conductive connectors **2130** may comprise, for example, ball grid array (BGA) connectors, solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, or the like. The conductive connectors **130** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the conductive connectors **130** are formed by initially forming a layer of solder through evaporation, electroplating, printing, solder transfer, ball placement, or the like. Once a layer of solder has been formed on the structure, a reflow may be performed in order to shape the material into the desired bump shapes. In another embodiment, the conductive connectors **130** comprise metal pillars (such as a copper pillar) formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like. The metal pillars may be solder free and have substantially vertical sidewalls. In some embodiments, a metal cap layer is formed on the top of the metal pillars. The metal cap layer may include nickel, tin, tin-lead, gold, silver, palladium, indium, nickel-palladium-gold, nickel-gold, the like, or a combination thereof and may be formed by a plating process. In some embodiments, the conductive connectors **130** comprise under-bump metallizations (UBMs) formed on the interconnect layers **120**.

[0072] FIGS. **23** through **26** illustrate intermediate steps in the formation of a package **200**, in accordance with some embodiments. The package **200** is similar to the package **100** described for FIG. **22**, except that the laser die **50** and the semiconductor die **150** are thinner, and the package **200** includes a support structure **207**. For example, in some embodiments, the laser die **50** and the semiconductor die **150** of the package **200** may have a thickness in the range of about 5 μm to about 50 μm, though other thicknesses are possible.

[0073] In FIG. **23**, the laser die **50** and the semiconductor die **150** are attached to an interposer structure **101**, in accordance with some embodiments. The interposer structure **101** may be similar

to the interposer structure **101** described previously for FIG. **18**. For example, the interposer structure **101** may include through vias **104** and an interconnect structure **110** that includes waveguides **112**. The laser die **50** and the semiconductor die **150** may be bonded to the interposer structure **101** using fusion bonding (e.g., metal-to-metal bonding and dielectric-to-dielectric bonding), which may be similar to that described previously for FIG. **20**. In this manner, the laser die **50** and semiconductor die **150** may be electrically and optically coupled to the interconnect structure **110** of the interposer structure **101**.
[0074] In FIG. **24**, an encapsulant **203** is formed on and around the laser die **50** and the semiconductor die **150**, in accordance with some embodiments. After formation, the encapsulant **203** encapsulates the laser die **50** and the semiconductor die **150**. The encapsulant **203** may be similar to the encapsulant **116** described previously for FIG. **21**. For example, the encapsulant **203** may be a molding compound, an epoxy, a polymer, a composite material, a dielectric material, an oxide (e.g., silicon oxide), or the like. In some embodiments, the encapsulant **203** is applied by deposition, spin-on, compression molding, transfer molding, or the like. The encapsulant **203** may be formed over the interconnect structure **110** such that the laser die **50** and the semiconductor die **150** are buried or covered. The encapsulant **203** may be applied in liquid or semi-liquid form and then subsequently cured.
[0075] Further in FIG. **24**, a planarization process is performed on the encapsulant **203** to expose the laser die **50** and the semiconductor die **150**, in accordance with some embodiments. Top surfaces of the laser die **50**, the semiconductor die **150**, and the encapsulant **203** may be substantially level or coplanar (within process variations) after performing the planarization process. In some cases, the planarization process may also remove upper portions of the laser die **50** and/or the semiconductor die **150**. The planarization process may comprise, for example, a chemical-mechanical polish (CMP) process, a grinding process, an etching process, or the like. In other embodiments, a planarization process is not performed.
[0076] In FIG. **25**, a bonding layer **205** is deposited over the encapsulant **203**, the laser die **50**, and the semiconductor die **150**, in accordance with some embodiments. The bonding layer **205** may comprise one or more materials suitable for dielectric-to-dielectric bonding, such a silicon oxide, silicon oxynitride, or the like. The bonding layer **205** may be deposited using a suitable process, such as CVD, PVD, ALD, or the like.
[0077] In FIG. **26**, a support structure **207** is attached to the bonding layer **205** to form the package **200**, in accordance with some embodiments. The support structure **207** may comprise one or more materials such as silicon, a dielectric (e.g., an oxide), metal, ceramic, plastic, the like, or a combination thereof. For example, the support structure **207** may be similar to the support substrate **38** or the support structure **162**. In some embodiments, the support structure **207** is bonded to the bonding layer **205** using fusion bonding, such as dielectric-to-dielectric bonding.
[0078] Further in FIG. **26**, interconnect layers **120** are formed on the back side of the substrate **102** as part of the interposer structure **101**, in accordance with some embodiments. The interconnect layers **120** may be similar to the interconnect layers **120** described previously for FIG. **22**. In some embodiments, the back side of the substrate **102** is thinned to expose the through vias **104** before forming the interconnect layers **120**. The interconnect layers **120** may be similar to the interconnect layers **106**, and may be formed using similar materials or techniques. For example, the interconnect layers **120** may comprise one or more layers of conductive features formed in one or more dielectric layers. In some embodiments, conductive connectors **130** are formed on the back side of the interposer structure **101** to form the package **200**, in accordance with some embodiments. The conductive connectors **130** may be similar to those described previously. In some cases, forming a package **200** as described may allow for the incorporation of thinner components or may result in an overall thinner package.
[0079] FIGS. **27**, **28**, and **29** illustrate example package structures **300**, **400**, and **500** that incorporate packages **100**, in accordance with some embodiments. The embodiments of FIGS. **27-**

29 are shown with packages **100**, but in other embodiments packages **200** may be used. The packages **100** shown in FIGS. **27-29** are similar to the package **100** shown in FIG. **22**, except that conductive connectors **130** may be different than shown in FIG. **22**. For example, the conductive connectors **130** may comprise conductive pads without solder bumps, in some cases. The embodiments shown in FIGS. **27-29** are intended as non-limiting examples, and other packages, package structures, or configurations thereof are possible.

[0080] FIG. **27** illustrates a package structure **300**, in accordance with some embodiments. The package structure **300** may be similar to an Integrated Fan-Out (InFO) package or the like, in some cases. In the package structure **300**, the package **100** is connected to a redistribution structure **302**. For example, the interposer structure **101** of the package **100** may be electrically connected to the redistribution structure **302** by conductive connectors **130**. The redistribution structure **302** includes metallization patterns formed in dielectric layers. The metallization patterns may also be referred to as redistribution layers or redistribution lines, and may comprise conductive lines, conductive vias, conductive pads, or the like. Conductive connectors **350** may be formed on the redistribution structure **302**, which may be similar to the conductive connectors **130** described previously. The package **100** may be encapsulated by an encapsulant **305**, and through vias **304** may be formed extending through the encapsulant to the redistribution structure **302**. A package component **310** may be connected to the through vias **304** by conductive connectors **306**, which may be solder balls or the like. The package component **310** may comprise one or more dies **308**A-B on a substrate **309**. The dies **308**A-B may be electrically connected to the through vias **304**. Other package components **310** are possible.

[0081] FIG. **28** illustrates a package structure **400**, in accordance with some embodiments. The package structure **400** may be similar to a flip chip package or the like, in some cases. In the package structure **400**, the package **100** is connected to a package substrate **402**. For example, the interposer structure **101** of the package **100** may be electrically connected to the package substrate **402** by conductive connectors **130**. The package substrate **402** may or may not include conductive routing, active devices, passive devices, or the like. Conductive connectors **450** may be formed on the package substrate **402**, which may be similar to the conductive connectors **130** described previously. A package component **410** may be connected to the package substrate **402** by conductive connectors **404**. The package component **410** may extend over the package **100**, in some cases. The conductive connectors **404** may be solder balls or the like. An underfill **405** may surround the package **100** and the conductive connectors **404**.

[0082] FIG. **29** illustrates a package structure **500**, in accordance with some embodiments. The package structure **500** may be similar to a Chip-on-Wafer-on-Substrate (CoWoS) structure or the like, in some cases. In the package structure **500**, multiple packages **100** are connected to an interposer **502**. For example, the packages **100** may be electrically connected to the interposer **502** by conductive connectors **130**. An underfill **508** may surround the conductive connectors **130**, in some cases. The interposer **502** may comprise a back side interconnect structure **504** and a front side interconnect structure **506** on a substrate. Through vias **505** may extend through the substrate to connect the interconnect structures **504** and **506**. Conductive connectors **550** may be formed on the back side interconnect structure **504**, which may be similar to the conductive connectors **130** described previously. An encapsulant **510** may be formed over the interposer **502** and the packages **100**, in some embodiments. These are examples, and other package structures incorporating packages **100**, **200**, or the like are possible.

[0083] The embodiments described herein can achieve advantages. Combining a laser die and a semiconductor die on an optical interposer can allow for optical and electrical signal communication within a package. The techniques described herein allow for the formation of a laser die that using processes and materials that are compatible with semiconductor fabrication. For example, the manufacturing of laser dies described herein allows for the use of noble metals or other materials in a manner that reduces the risk of contamination during subsequent processing of the laser dies into a package. The laser dies described herein utilize noble metals or other potential contaminants only for electrical contacts, and such materials are covered by electrical routing formed of more fab-compatible materials such as aluminum, copper, or the like. In this manner, the benefits of using noble metals in an electrical contact may be achieved without potentially contaminating exposure of the noble metals during subsequent processing. Accordingly, an optical package or the like may be formed without risk of contamination.

[0084] In some embodiments of the present disclosure, a method includes forming a laser diode structure including an active layer sandwiched between an n-type contact layer and a p-type contact layer; forming an n-type contact on the n-type contact layer, wherein the n-type contact includes a first noble metal; forming a p-type contact on the p-type contact layer, wherein the p-type contact includes a second noble metal; forming a conductive routing layer on the n-type contact and on the p-type contact, wherein the conductive routing layer is free of noble metals, wherein the conductive routing layer fully covers the n-type contact and the p-type contact; and forming a passivation layer over the conductive routing layer. In an embodiment, forming the n-type contact includes etching the n-type contact to form a recess and depositing the first noble metal into the recess. In an embodiment, the first noble metal is gold. In an embodiment, the second noble metal is platinum. In an embodiment, the conductive routing layer includes aluminum. In an embodiment, the method includes forming vias extending through the passivation layer to physically contact the conductive routing layer. In an embodiment, the method includes forming a conductive adhesion layer on the n-type contact and on the p-type contact, wherein the conductive routing layer is deposited on the conductive adhesion layer. In an embodiment, the method includes connecting the conductive routing layer to an optical interposer.

[0085] In some embodiments of the present disclosure, a method includes forming a first contact structure on a first contact layer of a laser diode, wherein the entire bottom surface of the first contact structure physically contacts a surface of the first contact layer; forming a second contact structure on a second contact layer of the laser diode, wherein the entire bottom surface of the second contact structure physically contacts a surface of the second contact layer; forming a first conductive layer on the first contact structure, wherein the first conductive layer is free of noble metals; and forming a second conductive layer on the second contact structure, wherein the second conductive layer is free of noble metals. In an embodiment, forming the first contact structure includes depositing a first stack of material layers, wherein at least one material layer is a noble metal. In an embodiment, the top material layer and the bottom material layer of the first stack are layers of gold. In an embodiment, forming the second contact structure includes depositing a second stack of material layers, wherein at least one material layer is a noble metal. In an embodiment, the first conductive layer includes copper. In an embodiment, the method includes recessing a portion of the first contact layer, wherein the first contact structure is formed in the recessed portion. In an embodiment, the method includes depositing a conductive adhesive layer on the first conductive layer and on the second conductive layer.

[0086] In some embodiments of the present disclosure, a device includes a laser die, which includes an n-type contact on an n-type contact layer; a multiple quantum well (MQW) layer over the n-type contact layer; a p-type contact on a p-type contact layer, wherein the p-type contact layer is over the MQW layer; and an aluminum routing layer on the n-type contact and on the p-type contact; an optical interposer bonded to the laser die, wherein the laser die is electrically and optically coupled to the optical interposer; and a semiconductor die bonded to the optical interposer, wherein the semiconductor die is electrically and optically coupled to the optical interposer. In an embodiment, the n-type contact includes a layer of gold, a layer of germanium, and a layer of nickel. In an embodiment, the p-type contact includes a layer of titanium and a layer of platinum. In an embodiment, a width of the p-type contact is less than a width of the p-type contact layer. In an embodiment, the n-type contact layer includes indium phosphide.

[0087] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming a laser diode structure comprising an active layer sandwiched between an n-type contact layer and a p-type contact layer; forming an n-type contact on the n-type contact layer, wherein the n-type contact comprises a first noble metal; forming a p-type contact on the p-type contact layer, wherein the p-type contact comprises a second noble metal; forming a conductive routing layer on the n-type contact and on the p-type contact, wherein the conductive routing layer is free of noble metals, wherein the conductive routing layer fully covers the n-type contact and the p-type contact; and forming a passivation layer over the conductive routing layer.
2. The method of claim 1, wherein forming the n-type contact comprises: etching the n-type contact to form a recess; and depositing the first noble metal into the recess.
3. The method of claim 1, wherein the first noble metal is gold.
4. The method of claim 1, wherein the second noble metal is platinum.
5. The method of claim 1, wherein the conductive routing layer comprises aluminum.
6. The method of claim 1 further comprising forming vias extending through the passivation layer to physically contact the conductive routing layer.
7. The method of claim 1 further comprising forming a conductive adhesion layer on the n-type contact and on the p-type contact, wherein the conductive routing layer is deposited on the conductive adhesion layer.
8. The method of claim 1 further comprising connecting the conductive routing layer to an optical interposer.
9. A method comprising: forming a first contact structure on a first contact layer of a laser diode, wherein the entire bottom surface of the first contact structure physically contacts a surface of the first contact layer; forming a second contact structure on a second contact layer of the laser diode, wherein the entire bottom surface of the second contact structure physically contacts a surface of the second contact layer; forming a first conductive layer on the first contact structure, wherein the first conductive layer is free of noble metals; and forming a second conductive layer on the second contact structure, wherein the second conductive layer is free of noble metals.
10. The method of claim 9, wherein forming the first contact structure comprises depositing a first stack of material layers, wherein at least one material layer is a noble metal.
11. The method of claim 10, wherein the top material layer and the bottom material layer of the first stack are layers of gold.
12. The method of claim 9, wherein forming the second contact structure comprises depositing a second stack of material layers, wherein at least one material layer is a noble metal.
13. The method of claim 9, wherein the first conductive layer comprises copper.
14. The method of claim 9 further comprising recessing a portion of the first contact layer, wherein the first contact structure is formed in the recessed portion.
15. The method of claim 9 further comprising depositing a conductive adhesive layer on the first conductive layer and on the second conductive layer.
16. A device comprising: a laser die, comprising: an n-type contact on an n-type contact layer; a multiple quantum well (MQW) layer over the n-type contact layer; a p-type contact on a p-type contact layer, wherein the p-type contact layer is over the MQW layer; and an aluminum routing

layer on the n-type contact and on the p-type contact; an optical interposer bonded to the laser die, wherein the laser die is electrically and optically coupled to the optical interposer; and a semiconductor die bonded to the optical interposer, wherein the semiconductor die is electrically and optically coupled to the optical interposer.

17. The device of claim 16, wherein the n-type contact comprises a layer of gold, a layer of germanium, and a layer of nickel.

18. The device of claim 16, wherein the p-type contact comprises a layer of titanium and a layer of platinum.

19. The device of claim 16, wherein a width of the p-type contact is less than a width of the p-type contact layer.

20. The device of claim 16, wherein the n-type contact layer comprises indium phosphide.